

Calibration-Aware Test-Time Adaptation for CLIP

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Abstract

Test-time adaptation (TTA) methods such as TDA [3] use only forward passes and a feature cache to recover much of the accuracy CLIP loses out of distribution, but a 2025 benchmark [5] showed that the gain comes with a hidden cost: TDA’s cache rewards confident predictions whether or not they are correct, so its accuracy gain arrives bundled with worse calibration (the benchmark reports average ECE rising 5.70% $\rightarrow$ 9.21% across its suite). We propose three training-free, label-free changes to TDA’s cache loop – margin-based cache admission, probabilistic fusion, and leave-one-out (LOO) temperature scaling – meant to recover calibration. Measured on five datasets (ViT-B/16, test split only), the three changes bundled together as specified *do not* win: mean accuracy drops from TDA’s 56.63% to 55.22% and mean ECE gets *worse*, from 11.24% to 24.12%. An ablation isolating each change shows why: two of the three contributions are actively harmful, and the third – LOO temperature scaling, run alone – is not. It matches the TDA-equivalent baseline’s accuracy exactly (56.60%, to 0.00 points, a structural guarantee of a monotonic rescaling) while cutting mean ECE from 11.28% to 8.54%, a 24% reduction toward zero-shot CLIP’s 5.57%. We report this honestly: the bundled method fails, one of its three components alone succeeds, and we explain the mechanism behind both outcomes.

1 Introduction

CLIP’s zero-shot accuracy is strong on natural-image classification but collapses on out-of-distribution domains: the benchmark that motivates this project reports 23.7% on EuroSAT and 15.7% on Aircraft [5]. Our own zero-shot measurements (Table 3) are higher – 42.06% and 23.85% – because we evaluate the CoOp test splits with a ViT-B/16 backbone rather than that paper’s exact setup; the collapse relative to in-distribution accuracy is the point, and it reproduces. Test-time adaptation (TTA) methods close much of this gap without any labels or backpropagation. TDA [3], in particular, keeps a small cache of features from test images the model is confident about and reuses that cache to refine later predictions – a forward-pass-only method with no training loop.

Accuracy, however, is not the only axis that matters for a system meant to be trusted. A model that is right 80% of the time but reports 95% confidence on every prediction is *worse* to deploy than one that is honest about its 80%, because a downstream user or pipeline has no way to tell a confident-and-correct prediction from a confident-and-wrong one. This is exactly the failure mode Sheng et al. [5] document in TDA: its cache rewards confident predictions with more confidence regardless of correctness, so accuracy gains arrive bundled with a calibration loss (average ECE rising 5.70% $\rightarrow$ 9.21% across their benchmark).

This project asks whether that calibration gap can be closed without retraining and without labels, by changing three seams of TDA’s cache loop:

1. **Margin-based cache admission.** Admit a test sample to the cache by the gap between its top-1 and top-2 class scores, instead of by prediction entropy, so the cache does not accumulate merely *loud* predictions.

2. **Probabilistic fusion.** Combine CLIP’s and the cache’s evidence as a convex mixture of probability distributions, instead of adding raw logits, so agreement between the two sources cannot manufacture confidence that neither source had on its own.
3. **Leave-one-out (LOO) temperature scaling.** Estimate the cache’s own accuracy without ground truth, via leave-one-out nearest-neighbour agreement among cached items, and pick a softmax temperature that makes the model’s mean confidence match that label-free estimate.

Each change is independently switchable from configuration, which is what makes the ablation in Section 5 possible – and the ablation is where this project’s real finding lives. **The three changes together do not improve calibration: they make it dramatically worse.** Run in isolation, however, one of the three delivers exactly the calibration improvement this project set out to find, at no accuracy cost. We report both results without softening either one.

2 Related Work

Training-free test-time adaptation for CLIP. TDA [3] adapts CLIP at test time with two small feature caches – a positive cache keyed by the model’s own predicted class, and a negative cache of low-confidence samples used as a correction term – built and consumed purely from forward passes, with no backpropagation and no labels. Each new test image is scored by CLIP and optionally admitted to the positive cache by an entropy threshold; the final logits are a sum of CLIP’s own logits and an α/β -weighted cache-logit term computed from the currently cached features. This project’s CalTDA, in `online_method/cal_tda.py`, subclasses TDA directly and reuses its cache-update and cache-logit machinery unmodified; the three contributions in Section 3 change only the admission rule, the CLIP/cache combination rule, and a final temperature.

The calibration cost of TTA. Sheng et al. [5] benchmark a broad set of TTA methods for vision-language models, including TDA, across the same class of few-shot recognition datasets used here, and report accuracy alongside expected calibration error (ECE). Their finding motivates this entire project: TTA’s forward-pass-only adaptation loop tends to raise accuracy while *also* raising ECE, because methods that reward and reuse confident predictions have no mechanism to distinguish confident-and-correct from confident-and-wrong. Their reported average ECE for TDA rises 5.70% $\rightarrow$ 9.21% on their suite even as accuracy improves. This project targets that specific gap using TDA as the adapted method and their framing of the problem, while running fully independent experiments (Section 4) rather than reusing their reported numbers.

Temperature scaling and ECE. Expected calibration error (ECE) buckets predictions by confidence and measures the gap between confidence and empirical accuracy within each bucket, weighted by bucket occupancy [1]. Temperature scaling divides logits by a single learned scalar T before the softmax. The operation is rank-preserving (it cannot change which class a model predicts) but reshapes confidence, and T is normally fit by minimizing negative log-likelihood on a labelled validation set. This project cannot use a labelled validation set – the spec’s no-ground-truth constraint applies throughout adaptation – so Section 3 replaces the usual supervised fit with a leave-one-out, label-free accuracy estimate computed from the method’s own cache.

3 Method

All three changes are implemented as switchable seams in `online_method/cal_tda.py`, whose CalTDA class subclasses upstream TDA’s runner class. With every flag off, CalTDA reduces exactly to upstream TDA – same caches, same per-dataset α/β hyperparameters, same additive

combination – which is the equivalence this project treats as a correctness invariant (verified at the unit-test level in `tests/test_cal_tda_runner.py` and, at the level of a full run, by the `none` row of the ablation in Table 4).

3.1 Contribution 1: margin-based cache admission

Upstream TDA offers every test sample to its positive cache; entropy is used only as an eviction key for samples already in the cache, not as an admission gate. This project replaces unconditional admission with a decisive-prediction gate. Given the logit vector z for a sample, let $z_{(1)}$ and $z_{(2)}$ be its two largest entries. The *margin* is

$$m = z_{(1)} - z_{(2)}, \quad (1)$$

and a sample is admitted to the positive cache only when m exceeds a fixed global threshold (`margin_threshold` = 1.0). Rationale: entropy conflates “the model is decisive” with “the model is decisive about a class it has seen a lot of evidence for”; the top-1/top-2 gap is a more direct decisiveness signal and does not require normalizing by the number of classes.

3.2 Contribution 2: probabilistic fusion

Upstream TDA finalizes a prediction as $z_{\text{clip}} + \alpha z_{\text{cache}}$, a sum of raw logits, and only then applies a softmax. This project instead fuses in probability space: given the CLIP logits z_{clip} and the cache logits z_{cache} , and a fusion weight $w \in [0, 1]$,

$$p = (1 - w) \sigma(z_{\text{clip}}) + w \sigma(z_{\text{cache}}), \quad (2)$$

where $\sigma(\cdot)$ is the softmax. Every run in this report uses $w = 0.5$. A convex combination of two distributions is bounded by the more confident of its two inputs, so two sources that happen to agree cannot combine into a confidence neither source had – exactly the guarantee additive logit-summing lacks, since summing before the softmax lets confident-and-agreeing logits amplify each other.

3.3 Contribution 3: leave-one-out temperature scaling

Standard temperature scaling fits a scalar T against a labeled validation set. No labels are available here, so T is instead chosen to match a label-free accuracy proxy computed from the method’s own positive cache. Let the cache hold M feature vectors together with the class each was cached under (the class the method itself predicted for that image, not a ground-truth label). For each cached item i , find its nearest neighbour among the other $M - 1$ cached items by cosine similarity and check whether the neighbour’s pseudo-label agrees with item i ’s own. The leave-one-out accuracy estimate is the fraction that agree:

$$\hat{a} = \frac{1}{M} \sum_{i=1}^M \mathbf{1}[y_{\text{nn}(i)} = y_i], \quad (3)$$

where $\text{nn}(i)$ is item i ’s leave-one-out nearest neighbour and y denotes the cache’s pseudo-labels. A well-calibrated classifier’s mean top-1 confidence should equal its accuracy, so the temperature is chosen to make the two match: writing $\bar{c}(T)$ for the mean top-1 softmax confidence over the run’s score history after dividing logits by T ,

$$T^* = \arg \min_T |\bar{c}(T) - \hat{a}|, \quad (4)$$

found by grid search rather than a closed form. The search grid is deliberately *smoothing-only*: its floor is $T = 1.0$ (the identity), so no candidate can sharpen a prediction, only soften it. This is a design constraint, not an oversight. Margin admission selectively caches high-margin, likely-correct samples, which biases $\hat{a}$ upward relative to the model’s true accuracy on the full test stream. An unconstrained search would respond to that inflated target by sharpening predictions, making overconfidence worse rather than better. Setting the grid’s floor at $T = 1.0$ means the search can only push confidence down or leave it alone, never manufacture more of it. (This rationale is the reason the floor is kept, not the reason it was first added: an initial single-dataset run without it showed exactly the failure mode described above, sharpening predictions and worsening ECE. The mechanism holds regardless of which reason came first.) Temperature scaling is a monotonic rescaling of the logits, so applying it never changes a prediction, only its reported confidence.

All three contributions are independently switchable. Each reads a boolean config flag (`use_margin_admission`, `use_prob_fusion`, `use_loo_temperature`), so all eight on/off combinations are runnable from configuration alone, which is the space Section 5’s ablation sweeps.

3.4 Code and attribution

This project is a fork of TomSheng21/`tta-vlm` [4], which supplies the evaluation framework used throughout: the CLIP wrapper, the dataset loaders, the entry points, and the `tda` and `clipzs` baselines every number in this report is compared against. The cache-update and cache-logit machinery that `cal_tda` inherits and reuses unmodified originates in `kdiaaa/tda` [2] (MIT-licensed) as ported into that fork. Our own contribution is `online_method/cal_tda.py` and the calibration, temperature-search, evaluation, analysis, and reporting code around it. The full source – including the scripts that regenerate every number, table, and figure in this report from the raw run records – is at https://github.com/macwave12/deep_project_calibration_aware_tta.

4 Experiments

4.1 Datasets and protocol

All numbers in this report are measured on the *test split only* of five standard few-shot recognition datasets, using the CoOp split files so counts match the benchmark literature exactly. Table 1 lists them:

Dataset	Test images	Classes
DTD	1,692	47
Oxford Flowers 102	2,463	102
Oxford Pets	3,669	37
EuroSAT	8,100	10
FGVC Aircraft	3,333	100

Table 1: The five evaluation datasets: test-split size and class count (CoOp splits).

Every method – zero-shot CLIP (`clipzs`), TDA (`tda`), and our method (`cal_tda`) – is evaluated one image at a time in a single online pass over the test split, with no repeated epochs and no access to any dataset’s training or validation split. Ground-truth labels are held out of every adaptation step: the cache, the margin gate, the fusion weight, and the LOO temperature search all operate on the model’s own predictions (“pseudo-labels”), never on a true label; labels are opened only once

per run, by the final scorer, to compute accuracy and ECE. The backbone is ViT-B/16 throughout, with batch size 1 and seed 0.

Calibration is measured with expected calibration error (ECE), computed with 20 equal-width bins by the single implementation in `utils/calibration.py` that scores every method identically.

4.2 Cache admission rates

Margin admission (Section 3, Contribution 1) uses one *global* threshold (`margin_threshold = 1.0`), deliberately not tuned per dataset – fitting it to each dataset’s own test-set margin distribution would be test-set tuning, which the no-labels protocol above is meant to rule out. Because the logits are $100\times$ a cosine similarity (CLIP’s `logit_scale` times the dot product of two unit vectors), a threshold of 1.0 asks for a top-1/top-2 cosine gap of only 0.01 – a quantity with no dataset-independent meaning. Measured admission rates on the full `cal_tda` run (all three contributions on) show what that threshold costs (Table 2):

Dataset	Classes	Zero-shot accuracy	Admission rate
Aircraft	100	23.85%	15.9%
EuroSAT	10	42.06%	26.5%
DTD	47	44.39%	53.0%
Flowers102	102	67.32%	65.7%
Pets	37	88.20%	81.9%

Table 2: Cache admission rates under the single global margin threshold (`margin_threshold = 1.0`), ordered by admission rate.

The gate ranges from admitting only 15.9% of samples on Aircraft – where the cache mostly starves – to 81.9% on Pets, where it is nearly a no-op and the method has quietly reverted to upstream TDA’s unconditional admission.

What varies with the gate is not class count. Flowers102 has the most classes of the five and the second-*highest* admission rate; EuroSAT has the fewest and the second-lowest. The ordering that does hold is the one shown in the table: across all five datasets, admission rate is rank-correlated with zero-shot CLIP’s accuracy, without exception. The mechanism is straightforward – where CLIP already separates the classes (pets, flowers), top-1/top-2 gaps are wide and the gate opens; where CLIP is lost (fine-grained aircraft variants, out-of-domain satellite imagery), every class looks similar to every other, margins collapse, and the gate closes. Five datasets cannot establish a law, even though a perfect rank match among five has roughly a 1% chance of arising by accident, so we state the ordering as the pattern our runs show rather than as a proven relationship.

That ordering is, however, the least useful behaviour a gate could have: it fills the cache where the cache is least needed and starves it precisely where adaptation had the most room to help. That mismatch, not the raw spread, is why margin admission, run alone, costs accuracy (56.60% $\rightarrow$ 53.67%, Table 4). We report it as a limitation of the fixed-threshold design (Section 6), not as a bug to be silently retuned away.

5 Results

5.1 Headline: the bundled method as specified

Table 3 is generated directly by `analysis/aggregate.py` from `results/summary.csv`; no number in it is hand-typed.

	Accuracy			ECE		
	clipzs	tda	cal_tda	clipzs	tda	cal_tda
Aircraft	23.85	24.27	23.97	5.44	15.51	8.65
DTD	44.39	47.04	45.63	8.27	17.41	17.72
EuroSAT	42.06	53.17	48.88	7.16	14.38	17.53
Flowers102	67.32	69.91	68.86	2.63	6.39	32.73
Pets	88.20	88.77	88.74	4.37	2.52	43.98

Table 3: Top-1 accuracy and ECE (% , 20 bins) on the five test splits.

Averaged over the five datasets, zero-shot CLIP reaches 53.16% accuracy at 5.57% ECE, TDA reaches 56.63% at 11.24% ECE, and `cal_tda` with all three contributions enabled – the method exactly as specified – reaches 55.22% at 24.12% ECE. **This is a clear failure of the bundled method.** Accuracy is lower than TDA’s on every single dataset (from 0.03 points on Pets up to 4.29 points on EuroSAT), and ECE is worse than TDA’s on four of the five datasets, dramatically so on Flowers102 (32.73% vs. 6.39%) and Pets (43.98% vs. 2.52%). Aircraft is the sole dataset where the bundle’s ECE number looks better than TDA’s (8.65% vs. 15.51%) – Section 6 shows this is not real correction, only the same underconfidence failure landing closer to correct by accident on this one low-accuracy dataset. We do not reframe this as a partial success: on the spec’s own success criterion (accuracy $\geq$ TDA *and* ECE $<$ TDA, on at least 3 of 5 datasets), the bundled method meets the bar on **0 of 5** datasets.

Figure 1 plots the same comparison directly: our success criterion is a point as far right as TDA (equal accuracy) but as far down as zero-shot CLIP (honest confidence). The bundled method’s green star sits left of TDA’s orange square on every dataset (lower accuracy) and above it on four of the five (higher ECE) – Aircraft is the one dataset where the star sits below the square, matching the exception already noted above.

5.2 Reliability diagrams

Figure 2 plots, per dataset, the confidence-vs-accuracy curve for each of the three headline methods plus the ablation’s `temp`-only variant (a curve on the dashed identity line is perfectly calibrated). The grid makes the aggregate ECE numbers in Table 3 legible per bin, and the direction of the failure is visible directly: on every dataset, `cal_tda`’s curve sits *above* the identity line – accuracy exceeds stated confidence, not the reverse. In the five `cal_tda` reliability records the highest occupied confidence bin tops out at 0.70 on EuroSAT and at 0.60 or below on the other four, so there is no high-confidence region to be confidently wrong in; the method is instead severely *underconfident*, most visibly on Pets, where 88.74% accuracy is reported at only 44.77% mean confidence. Section 6 explains the mechanism, including why Aircraft – the one dataset where the bundle’s ECE looked like an improvement over TDA in Table 3 – is not a case of the method correcting itself, but the smallest instance of the same underconfidence effect.

5.3 The ablation

Table 4 (generated by `analysis/make_ablation_table.py` from a 40-run sweep – all eight on/off combinations of the three contributions, on all five datasets) isolates which of the three changes are responsible for the failure above.

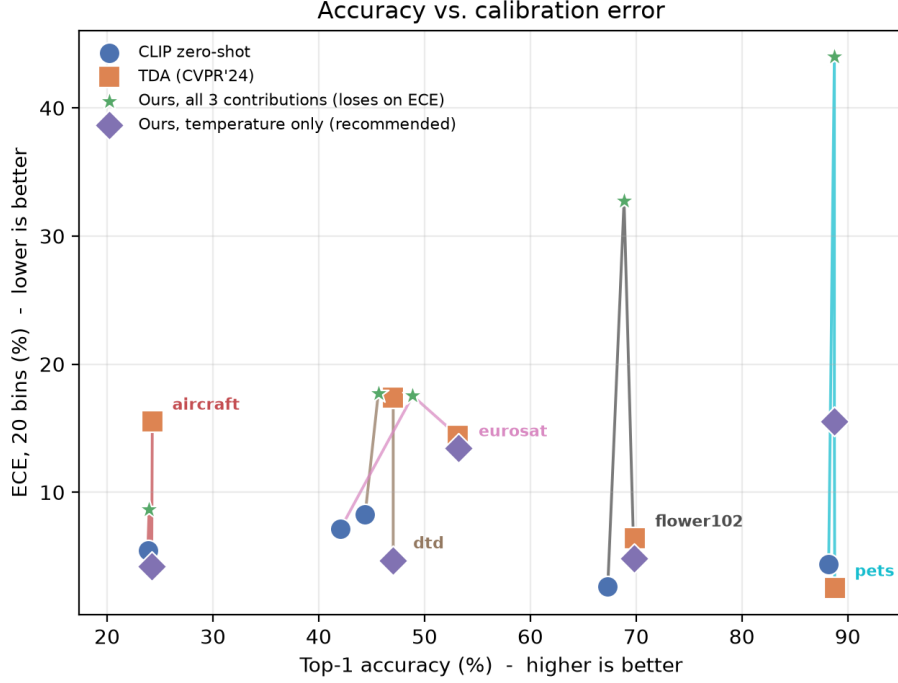


Figure 1: Accuracy vs. ECE, all five datasets. The purple diamond (LOO temperature only, Section 5.3) is the configuration this report ultimately recommends, plotted alongside the three headline methods. Marker shape and colour identify the method (see legend); each dataset’s four points are joined by a line in that dataset’s own colour, labelled at the line’s right-hand end. Lines run left to right by accuracy; they mark grouping only, not a progression.

	Accuracy	ECE
none: TDA-equivalent (all flags off)	56.60	11.28
margin: + margin admission	53.67	18.06
fusion: + probabilistic fusion	55.60	24.54
temp: + LOO temperature	56.60	8.54
margin_fusion: + margin + fusion	55.22	23.94
margin_temp: + margin + temperature	53.67	19.39
fusion_temp: + fusion + temperature	55.60	24.78
all: all three (ours)	55.22	24.12

Table 4: Ablation: mean accuracy and ECE (%) across the five datasets.

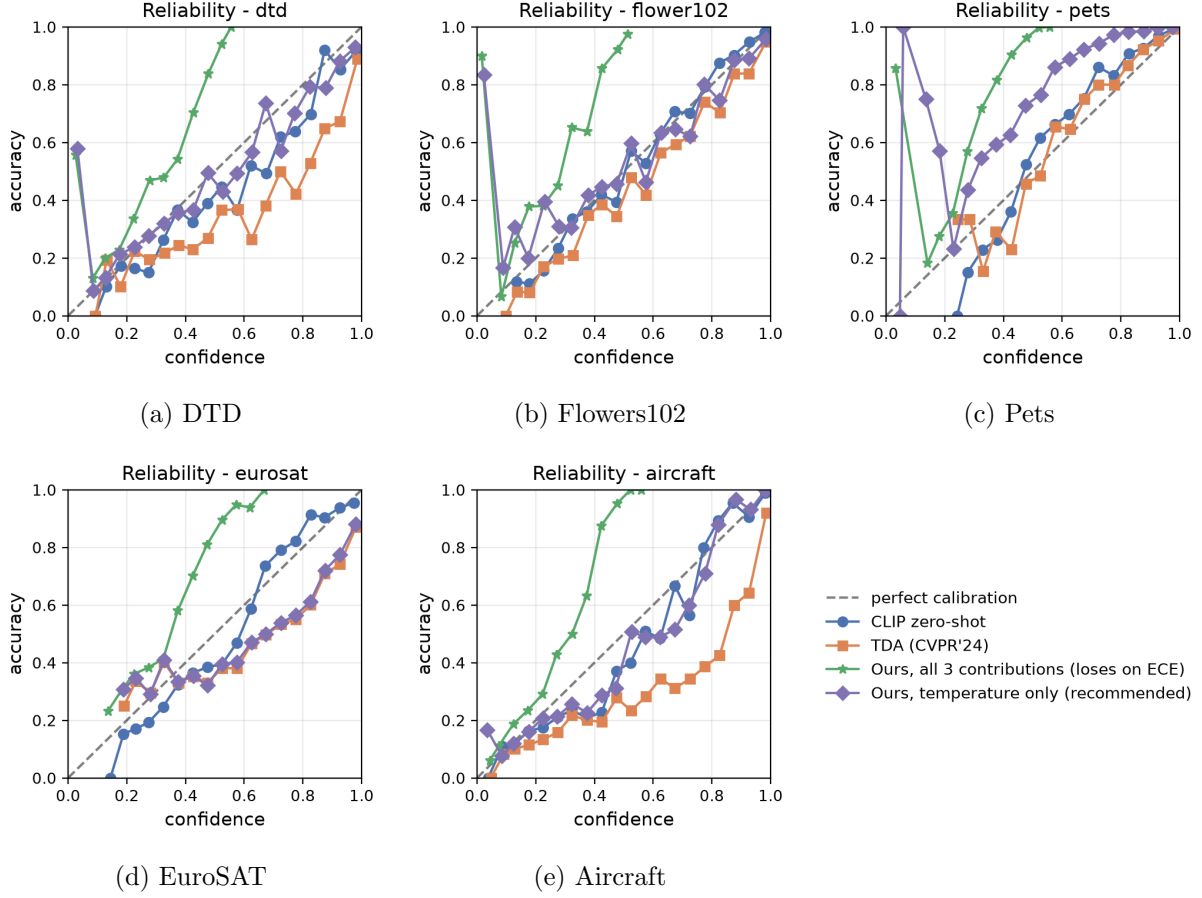


Figure 2: Reliability diagrams, all five datasets: zero-shot CLIP, TDA, `cal_tda` (all three contributions), and the ablation’s `temp`-only variant. All five panels share the legend at bottom right; a curve on the dashed identity line is perfectly calibrated.

Contribution 3 (LOO temperature scaling) alone succeeds. The ablation’s own `none` baseline is `cal_tda` with all three flags off, run inside the same 40-run sweep; it is the correct same-code-path comparison, since `temp` and `none` differ by exactly one config flag. Against it, `temp` matches accuracy *exactly*, to 0.00 points, on all five datasets – a structural guarantee, since temperature scaling is a monotonic rescaling of the logits and cannot change an argmax. It cuts mean ECE from 11.28% to 8.54%, the best mean ECE of all eight ablation variants, including `all` (24.12%). The per-dataset breakdown is in Table 5:

Dataset	<code>none</code> ECE	<code>temp</code> ECE	Change
Aircraft	15.56	4.21	−11.35
DTD	17.41	4.68	−12.73
EuroSAT	14.43	13.43	−1.00
Flowers102	6.51	4.85	−1.66
Pets	2.48	15.54	+13.06 (worse)

Table 5: Per-dataset ECE (%) for the ablation’s `none` and `temp` variants. Pets is the one reversal.

This is not a uniform win: on Pets, where the `none` baseline was already very well calibrated (2.48% ECE), LOO temperature scaling makes calibration substantially *worse*. We discuss the mechanism in Section 6 rather than average it away.

Two counts, and why they differ. Against the ablation’s `none` baseline, `temp` meets the spec’s success criterion on **4 of 5** datasets (all but Pets). Against the separately run headline `tda` rows in Table 3, the count is lower – **2 of 5** – because `none` and `tda` are two different process invocations of what should be identical logic, and in practice differ by 0–4 flipped predictions per dataset out of 1,692–8,100 samples. The cause is not a logic difference: upstream `tda` accumulates its final logits in fp16, while `cal_tda` (even with every flag off) upcasts to fp32 before its final step, and at CLIP’s logit magnitudes that rounding difference is worth roughly one unit in the last place – enough, on borderline-margin predictions, to flip an argmax once every few hundred to a few thousand samples. Both counts are correct; they answer different questions. The 4-of-5 count answers whether temperature scaling, holding everything else fixed, helps calibration without moving accuracy – yes, on 4 of 5. The 2-of-5 count answers whether a `temp`-only run, as a real separate process, beats the specific `tda` numbers quoted in Table 3 – only on 2 of 5, because of run-to-run floating-point drift that is two orders of magnitude smaller than any effect discussed above.

Contributions 1 and 2 are the ones actually responsible for the bundle’s failure. Margin admission alone *lowers* mean accuracy (56.60% → 53.67%) and raises mean ECE (11.28% → 18.06%): its selectivity throws away cache signal the unfiltered cache was using correctly on some datasets. Probabilistic fusion alone is the worst single component: it raises mean ECE to 24.54%, driven by blowups on Flowers102 (32.46%) and Pets (43.77%) specifically, though it happens to help Aircraft (15.56% → 9.16%) – the reason “all three” still beats plain TDA’s ECE on that one dataset. Neither contribution helps consistently even in isolation; both actively hurt the mean.

Cache admission rates, which explain part of why margin admission behaves so differently across datasets, are reported in Section 4.2.

6 Discussion

6.1 Where it worked, and why

LOO temperature scaling alone is a real, if not universal, calibration improvement. The mechanism is direct: it always smooths the model’s confidence toward a single label-free accuracy estimate computed from the cache. Where TDA’s cache was badly overconfident relative to its own accuracy – DTD (17.41% $\rightarrow$ 4.68% ECE) and Aircraft (15.56% $\rightarrow$ 4.21%, same-code-path **none-vs-temp** comparison) chief among them – pulling confidence down toward the estimate is exactly the correction needed, and the improvement is large.

6.2 Where it did not, and why

The same mechanism explains the Pets reversal. On Pets, TDA’s cache was already close to perfectly calibrated (2.52% ECE for the headline **tda** run, 2.48% for the ablation’s **none** baseline) – the model’s stated confidence already matched its accuracy. LOO temperature scaling has no notion of “already well calibrated”; it always smooths toward the LOO estimate regardless of how good the starting point was, so with no overconfidence to correct, it manufactures underconfidence instead (ECE rises to 15.54% under **temp**, or as high as 43.98% under the full bundle). A method that only ever pushes confidence toward one estimate, with no mechanism to detect when the starting point does not need correcting, will help wherever that estimate happens to be closer to the truth than the model’s own confidence, and hurt wherever the estimate is not.

Margin admission and probabilistic fusion fail for related but distinct reasons. Margin admission’s single global threshold (**margin.threshold** = 1.0) is not portable: measured admission rates (Section 4.2) run from 15.9% on Aircraft to 81.9% on Pets, and they track zero-shot CLIP’s accuracy rather than class count. On Aircraft the cache mostly starves; on Pets the gate is nearly a no-op. The gate is therefore most selective exactly where the cache had the most to contribute. Neither extreme is what a per-dataset-tuned gate would look like, and, as Section 5 shows, its selectivity costs accuracy outright (56.60% $\rightarrow$ 53.67% under margin admission alone).

Probabilistic fusion, meanwhile, is the single worst component in the ablation (mean ECE 24.54%) – but not for the reason it first appears. **compute_cache_logits** (inherited unmodified from upstream TDA) returns, for each class, a sum of $\exp(-\beta(1 - \text{affinity}))$ terms over whatever cached items belong to that class – an unnormalized, comparatively small-magnitude quantity, since affinity is a cosine similarity in $[-1, 1]$ and only the handful of classes actually represented in a small cache get a nonzero score at all. $\sigma(z_{\text{cache}})$ is therefore close to uniform relative to $\sigma(z_{\text{clip}})$, which is sharply peaked because CLIP’s own logits are already scaled by its learned logit scale (≈ 100). Averaging a near-uniform distribution into a near-one-hot one at $w = 0.5$ (Eq. (2)) caps top-1 probability well below what CLIP alone would report, *regardless of how much evidence actually supports the prediction* – fusion does not need both sources to agree on a wrong answer to produce this effect; it produces the effect on every sample, right or wrong alike. This single fact accounts for everything the reliability diagrams show (Section 5): no **cal_tda** reliability record on any dataset reaches even 0.70 mean confidence in its highest-occupied bin, mean confidence sits 8.7–44 points below accuracy on every dataset, and the size of that gap tracks accuracy itself: it is largest on the highest-accuracy dataset (Pets, 88.74% accuracy capped near 44.77% confidence) and smallest on the lowest-accuracy one (Aircraft, 23.97% accuracy capped near 15.32%). That is what a fixed confidence ceiling does – it costs a high-accuracy dataset far more calibration error than a low-accuracy one. Fusion’s ECE is therefore not overconfidence from agreeing wrong answers; it is underconfidence manufactured by capping every prediction’s confidence near one fixed point, independent of correctness.

This also explains Aircraft’s apparent improvement in Table 3 without invoking any real correction. The **none** baseline on Aircraft is overconfident to begin with (mean confidence 39.77% against 24.21% accuracy, ECE 15.56%) – ordinary TDA-style overconfidence. Adding fusion does not fix that overconfidence; it overshoots straight past it, capping confidence at 15.2–15.4% against 23.97–24.42% accuracy (ECE 8.65–9.20%). The resulting ECE is smaller purely because Aircraft’s true accuracy (23.97%) happens to sit close to fusion’s confidence ceiling on this particular dataset – an accident of where the ceiling landed relative to Aircraft’s accuracy, not evidence that fusion diagnosed and corrected **none**’s overconfidence. On every other dataset, where accuracy is higher, the same ceiling produces a much larger gap in the other direction, which is why fusion is the worst component overall despite this one favourable-looking case.

6.3 The $\text{margin} \times \text{temperature}$ interaction, and its limits

An earlier single-dataset (DTD) observation suggested a clean interaction, and the five-dataset runs bear it out: once margin admission and fusion are both active, the LOO temperature search pins at its grid boundary $T = 1.0$ on 98–100% of samples on every dataset, meaning the search wants to sharpen further than the smoothing-only grid allows and Contribution 3 becomes effectively inert on top of Contributions 1 and 2 – consistent with the ECE difference between **margin_fusion** and **all** being tiny everywhere (0.00–0.36 points). Testing that story against temperature on top of margin *alone* (without fusion in between) shows it does not generalize cleanly: pinning drops to 75–95% on four datasets, with a correspondingly larger ECE effect (DTD improves by 2.81 points, Flowers102 by 2.25), and **Pets breaks the pattern outright** – its **margin_temp** run pins only 0.2% of samples at the boundary, the opposite of inert, and it is exactly on Pets, already well calibrated under margin alone (2.53% ECE), that adding temperature does the most damage of any margin-plus-temperature run (2.53% $\rightarrow$ 14.79%; the larger temperature-induced regression in the ablation is Pets without margin, Table 5: 2.48% $\rightarrow$ 15.54%). The single-dataset story pointed to one mechanism; five datasets show it holds strongly with fusion present, holds more weakly for margin alone, and inverts on the one dataset a single-dataset study would have missed. We report this as a caution about generalizing a mechanism from one dataset, not as a discarded finding – the underlying explanation (temperature is inert exactly when the search is already pinned against its floor) still holds; what changes across datasets is how often that precondition is met.

6.4 Accuracy/calibration trade-off, stated plainly

None of the three contributions, alone or combined, is a free lunch. Contribution 3 is the one change in this project that costs nothing in accuracy (it structurally cannot, being a monotonic rescaling) and buys a real mean calibration improvement – but even that improvement is not uniform, and reverses on the one dataset that was already well calibrated. Contributions 1 and 2 cost accuracy, calibration, or both, on average, and the one apparent exception – fusion’s lower ECE on Aircraft – is not a benefit either, once its mechanism is understood: it is the same confidence-capping failure landing, by the accident of that one dataset’s low accuracy, closer to correct than the overconfidence it replaced. The honest summary of this project’s central finding is that a training-free, label-free calibration fix is achievable, but it is one targeted change (temperature scaling on a label-free accuracy estimate), not the three-part bundle we set out to build.

7 Conclusion

This project set out to undo TDA’s calibration loss with three training-free, label-free changes to its cache loop, and measured whether that worked rather than assuming it. It did not: bundled together as specified, the three contributions meet the project’s own success criterion on 0 of 5 datasets, trading away both accuracy and calibration relative to plain TDA. But the ablation that made this failure legible also isolated a real success inside it – leave-one-out temperature scaling, run alone, matches TDA-equivalent accuracy exactly while cutting mean ECE by 24% toward zero-shot CLIP’s level, and meets the success criterion on 4 of 5 datasets, for a structural reason (a monotonic rescaling cannot move accuracy) that makes the result as close to guaranteed-safe as a calibration intervention gets. Margin admission and probabilistic fusion, the other two contributions, are the ones actually responsible for the bundle’s failure, and neither helps consistently even in isolation. The finding this report stands behind is therefore narrower and more specific than the one we set out to demonstrate: not “three changes fix TDA’s calibration,” but “one of three does, and we now know which, and why the other two do not.”

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